

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-1/2 EXAMINATION – WINTER 2021****Subject Code:3110011****Date:22/03/2022****Subject Name:Physics****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Define Hooke's law, stress and strain.	03
	(b) Explain various properties of LASER beam.	04
	(c) Describe Stress-Strain diagram in detail.	07
Q.2	(a) Define wave motion. Discuss different types of waves.	03
	(b) (i) A cinema hall has a volume of $9,500 \text{ m}^3$. What should be the total absorption in the hall if the reverberation time of 1.7 s is to be maintained?	02
	(ii) An ultrasonic source of 0.075 MHz sends down a pulse towards the seabed, which returns after 0.95 s. The velocity of ultrasound in sea water is 1800 m/s. Calculate the depth of the sea and wavelength of pulse.	02
	(c) Discuss in detail the different types of elasticity. List different factors affecting elasticity.	07
	OR	
	(c) What are the factors affecting acoustics of the building and give their remedies.	07
Q.3	(a) A brass bar having a cross-section of 1 cm^2 is supported on two knife-edges 1.5 m apart. A load of 2 kg at the center of the bar depresses that point by 2.75 mm. What is the Young's Modulus for brass?	03
	(b) Describe viscosity. How the comparison of viscosities of two liquids can be done?	04
	(c) Establish the relation between Einstein's coefficients.	07
	OR	
Q.3	(a) In the acoustic grating experiment, the wavelength of light transmitted through a liquid is $5970 \text{ \AA}$. The 1 st order angle of diffraction is 0.195° . Calculate the velocity of ultrasound in the liquid having frequency 2.7 MHz.	03
	(b) Expand SQUID. How is it formed and give its applications.	04
	(c) Explain in detail the production of ultrasonic waves through piezoelectric oscillator method.	07
Q.4	(a) Calculate the frequency to which piezoelectric oscillator circuit should be tuned so that a piezoelectric crystal of thickness 0.2 cm vibrates in its fundamental mode to generate ultrasonic waves. Young's modulus is 80 Gpa and density of material is 2654 kg/m^3	03
	(b) Define Cooper pair. Explain BCS theory for superconductors.	04
	(c) What are the applications of Ultrasound? Discuss them in detail.	07

OR

- Q.4** **(a)** The critical current density equal to $1.71 \times 10^8 \text{ A/m}^2$ is required to change a superconducting wire of radius 0.5 mm at 4.18 K. If the critical temperature of the material is 7.5 K, calculate the maximum value of the critical magnetic field. **03**
- (b)** Explain NDT with it's objectives. **04**
- (c)** Explain in detail construction and working of He-Ne Gas LASER with necessary schematic and energy level diagrams. **07**
- Q.5** **(a)** A voltage of $6.7 \mu\text{V}$ is applied across a Josephson junction. What is the frequency of the radiation emitted by the junction in GHz? **03**
- Planck's constant = $6.626 \times 10^{-34} \text{ J.s}$
- (b)** Define Optical resonator, life time, metastable state and pumping mechanism for LASER. **04**
- (c)** Define Superconductor. Discuss it's properties in detail. **07**
- OR**
- Q.5** **(a)** A hall has a volume of 2790 m^3 . It's total absorption is equivalent to 98.80 m^2 of open window. What will be the effect on reverberation time if the audience fill the hall and thereby increase the absorption by another 98.80 m^2 of open window. **03**
- (b)** What is ultrasonic waves? Give properties and detection methods for ultrasonics. **04**
- (c)** Give applications of LASER in various fields in detail. **07**
